

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Fall

Year : 2022
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Microprocessor along with its applications. Differentiate Harvard and Von-Neumann architecture on the basis of storage. 7
b) What are the types of instruction depending upon word size? Explain different type of addressing modes of 8085 microprocessor. 8
2. a) Write an 8085 program to find smallest number among array of numbers. 8
b) Define T-state. Draw the labelled timing diagram of the instruction LXI H, 5051H. 7
3. a) Design an Addressing Decoding circuit to interface 16KB ROM chips with 8085 microprocessors. 8
b) Draw a well labelled architecture of 8086 microprocessor and discuss about BIU and EU. 7
4. a) Explain about IVT of 8086. 7
b) What are serial and parallel communication? Explain basics of RS-232. 8
5. a) Explain the working of DMA. Mention its advantages over programmed and interrupt driven I/O. 8
b) Write an ALP in 8086 to display POKHARA UNIVERSITY without using 09h function. 7
6. a) What is the Asynchronous serial data communication? Compare serial and parallel communication. Draw the Block diagram of 8251A USART and explain functionally in detail. 8
b) What is the importance of DAA instruction? Explain it with an appropriate example along with the necessary calculations. 7
7. Write short notes on: (Any two)
a) Memory mapped I/O vs I/O mapped I/O
b) Address Bus, Data Bus and Control Bus
c) Flags in 8086